



Data Mining I Project:

Analyzing Board Games Data

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1 Introduction

The BoardGame dataset provides comprehensive information about board games, capturing data such as year of publication, complexity ratings (both manufacturer and community-assigned), recommended number of players, age suitability, language requirement, and playtime estimates. It also contains indicators of popularity and engagement, such as the number of users who own, wish to own, or have rated each game, along with community-driven metrics like recommended player counts, comments, and category rankings. This combination of features makes the dataset a valuable resource for analyzing trends in game design, community preferences, and the relationship between gameplay characteristics and user engagement.

2 Data Understanding and Preparation

2.1 Data Semantics

2.2 Distribution of the Variables and Statistics

In this subsection, we examine the distribution of the variables in the dataset. The variable `YearPublished` spans a very wide time range, from approximately 3500 BC to 2021. Its distribution is highly right-skewed, with a sharp increase in publications starting from the early 20th century and a peak in the 2000–2020 period. Two variables describe game complexity: `GameWeight` (manufacturer-declared) and `ComWeight` (community-evaluated). Both distributions show a concentration of default values equal to zero, indicating missing or unreported complexity. For `ComWeight`, a few values exceed the theoretical upper bound of 5.0. The variable `ComAgeRec` presents many missing values (5530) and several decimal entries; the raw distribution shows irregular patterns due to inconsistent reporting. The variables `MinPlayers` and `MaxPlayers` include default zeros that do not represent valid player counts. Their distributions confirm that the majority of games are designed for 2–4 players. The popularity-related variables (`NumOwned`, `NumWant`, `NumWish`, `NumWeightVotes`) are all extremely right-skewed, with most titles receiving very few interactions and only a limited number of highly popular games dominating the upper tail. Finally, the playtime variables (`MfgPlaytime`, `ComMinPlaytime`, `ComMaxPlaytime`) display large variability and heavy right skew. A significant portion of observations contains default zeros, especially in manufacturer-reported durations.

2.3 Assessing Data Quality

A rigorous assessment of data quality is essential before proceeding with the subsequent phases of the analysis. In this section, we describe the issues encountered during the initial inspection of the BoardGameGeek dataset and the corresponding preprocessing steps carried out to ensure consistency, interpretability, and

suitability for statistical modeling.

2.3.1 Data Quality

Several variables were identified as unsuitable for quantitative analysis due to their nature, semantic ambiguity, or lack of informative content. Following standard data preparation practices, these attributes were removed from the working dataset:

- **BGGId**: a unique game identifier with no predictive or descriptive use in a statistical context.
- **Description**: free-text narrative of the game; excluded as the project does not involve text mining.
- **ImagePath**: contains only file paths to game images and provides no analytical value.
- **GoodPlayers** and **BestPlayers**: both variables contain a large proportion of missing and inconsistent values, rendering them unreliable.
- **NumComments**: this attribute contains only zeros in the entire dataset, providing no information for modeling.

Removing these variables reduced dimensionality without causing any loss of relevant information for the purposes of our analysis.

2.3.2 Missing Values

A substantial number of attributes contained missing entries or default placeholder values (typically the value 0, used to indicate the absence of reported information rather than a true measurement). The imputation strategy adopted for each variable was tailored to its role, distribution, and measurement scale.

Name. This column is repetitive. 'Name' has repeated 355 times. To resolve this we have two solutions either (A) to obtain a keyword from 'Description' column and append it with Name or (B) using 'YearPublished' with name and if year is same adding a Suffix 1,2,...N. To reduce complexities we are using 'YearPublished' as it has no missing values.

LanguageEase.

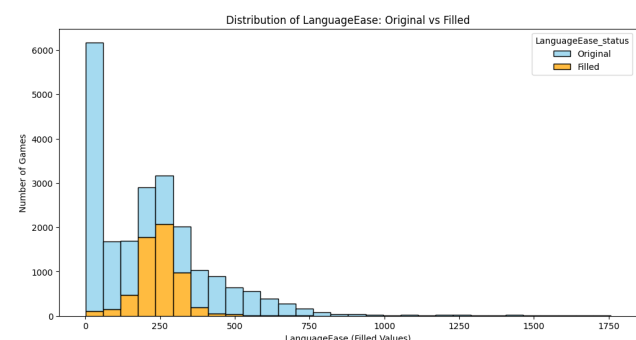


Fig. 1: LanguageEase missing values filled using mean.

The column contained 5891 missing rows. To fill these, we used the group-mean of a few numerical features which, according to correlation, had some percentage of relation. Missing values were filled as shown in the figure.

YearPublished. The publication year ranges from 3500 B.C. to 2021. Initially we found no Missing Values but seeing different characteristics this feature actually contains 193 zero values. The feature when passed to Pandas was kept as 'String' hence finding 0 or NA resulted nothing initially.

Since this is a year, time-based reference not relative to any feature, running relation-based Algorithm will not help. Grouped Median can also be vague as we cannot group YearPublish with any other numerical feature. We used a combined approach to fill these values, we first looked into 'Family' of missing YearPublished.

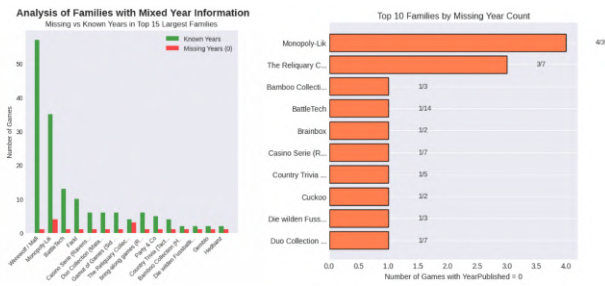


Fig. 2: Known and Missing YearPublished wrt Family

For games with missing YearPublished within a family, we computed median of the available years for that family and filled accordingly. For remaining missing values where no family existed, we used global median (we ignored the old or rare years to prevent distortion of median by high values). This approach helped in keeping diversity of the dataset.

Almost 29 values were filled using Family Median and 164 values filled using Global Median.

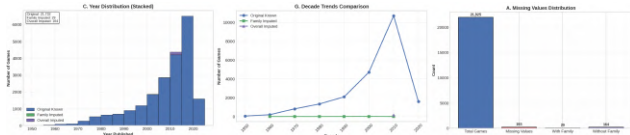


Fig. 3: Distribution of Family-based Median Filling and Global Median

GameWeight and ComWeight. Both GameWeight and ComWeight contained 506 default values equal to zero. We tried Grouped Medians and Linear Regression to fill these values. The results were mostly similar; to avoid unnecessary complexity, the missing values were filled using Grouped Medians.

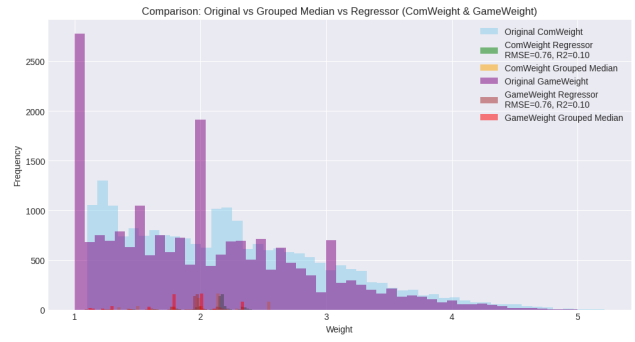


Fig. 4: GameWeight and ComWeight using Regression and Grouped Median.

NumWeightVotes. This feature represents number of community members who voted for a particular game. To fill missing values following strategies were considered:

- A) The feature is related to ComWeight, first we analysed if there exists any identical ComWeight values where NumWeightVote is 0, so we can directly fill missing entries from similar rows. However, identical ComWeight values were detected but with different NumWeightVote values. Which tells that same ComWeight can exist with different rating distributions.

Rows with ComWeight (which also exist when NumWeightVotes are missing): 14

	Name	ComWeight	NumWeightVotes
0	Titty Grab	1.2869	2
1	Word Slam	1.2869	18
2	Catch Phrase!	1.3380	162
3	Wheel of Fortune	1.3448	38
4	Tadaaam!	1.4181	13
5	Dilbert: Corporate Shuffle	1.4181	60
6	Konexi	1.5057	8
7	Black Stories 2	1.5057	43
8	Monopoly: Stranger Things	1.7827	3
9	Woodlands	2.0065	9
10	Pieces of Eight	2.0065	23
11	The Mob	2.0065	24
12	Oracle	2.2997	1
13	Scanderoon	2.2997	6

Fig. 5: ComWeight rows showing varying NumWeightVotes for non-missing NumWeightVotes

- B) The feature is 0.9 correlative to 'NumUserRatings'. And only 506 missing values exist. We tried Linear Regression, KNN Classification and Decision Tree to see best possible substitute with all numerical features and with only 'NumUserRatings', 'GameWeight' and 'ComWeight'.

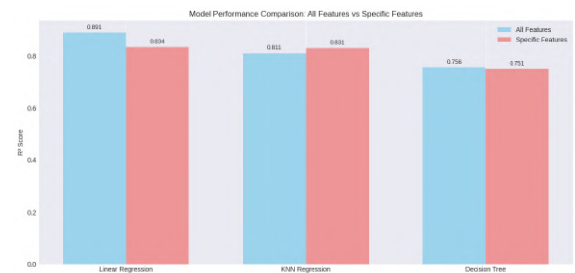


Fig. 6: Correlation analysis for NumWeightVotes

Dataset was split into NON-NA train(70%) and validation(30%). Results from validation for LinearRegression had better R^2 , hence values were filled using Linear Regression.

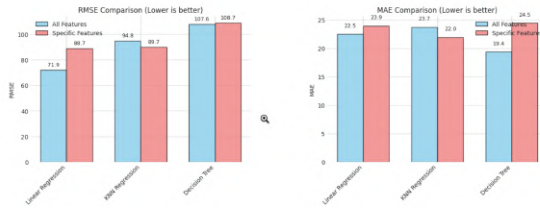


Fig. 7: RMSE and MAE for all three algorithms wrt limited and all features

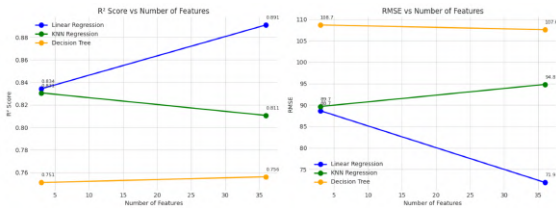


Fig. 8: Impact of Features on Performance

In Figure 8, we can see the influence of features on Performance Metrics of all three algorithms.

MfgAgeRec. Manufacturer Age feature contained 6.0% missing values. We first tried to check Mean, Mode and Median of this feature within rows as well as combined with other correlated features such as ComAgeRec, GameWeight, Cat:Childrens, Cat:War, Cat:Strategy.

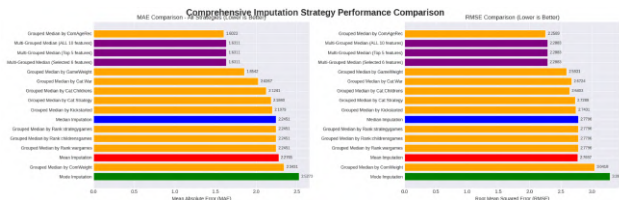


Fig. 9: MAE and RMSE wrt Normal Mode Median and Grouped

Above MAE and RMSE were calculated by predicting missing values of MfgAgeRec on validation non-zero dataset. So far the better results yielded were of Group Median with ComAgeRec.

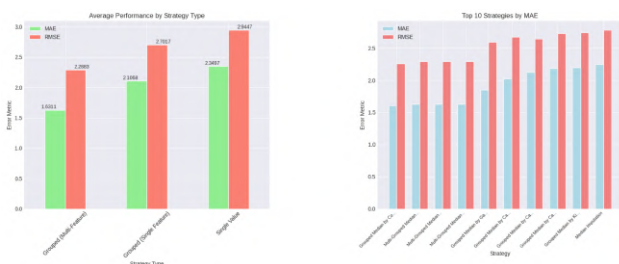


Fig. 10: Comparison of MAE and RMSE throughout various strategies

Before checking Classification Algorithms, we tried looking for Patterns to fill Missing Values, we used **FP-Growth** to see potential association-rules.

For FP-Growth, we first tried to discretize the MfgAgeRec data with different strategies.

A) Custom Bins by seeing Data Distribution:

We tried to divide non-zero data into four bins according to age groups, based on the observed data distribution but results were biased for Children (6-12) Bin. Binning to Five Bins according to formula: $\sigma = \frac{X_{\max} - X_{\min}}{K}$. Substituting the values: $\sigma = \frac{25-0}{5} = 5$ was not any better.

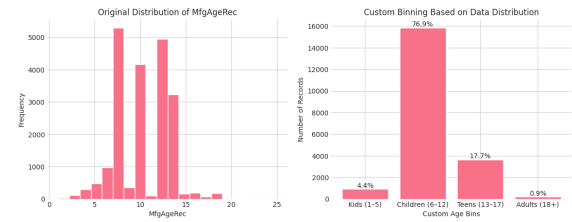


Fig. 11: Original Distribution and Custom Binning Distribution

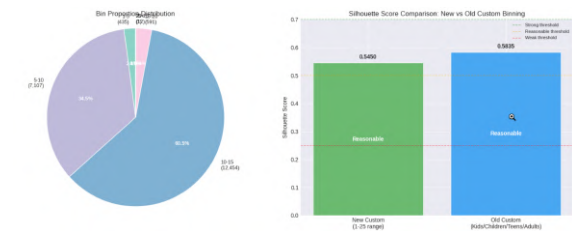


Fig. 12: Comparison of 4 and 5 Bins Binning

Clearly the dataset up to this part is highly skewed with a lot of games falling in 6-10 and 10-15 age (Children Bins) in MfgAgeRec. Applying Pattern Mining here resulted in associate rules for dominant bin only. At **Confidence** of 0.60, 665 age prediction rules were generated, almost all targeting Age_Children Frequent Patterns.



Fig. 13: Comparison of 4 and 5 Bins Binning

Number of Bins were increased to 6 in order to see if we can format rules for other Age Groups. This approach generated richer set of association rules, covering more age groups. Still majority of rules targeted Older_Children and Younger_Children (wrt Bins) but minority of rules included Older_Teen and Adult as well.

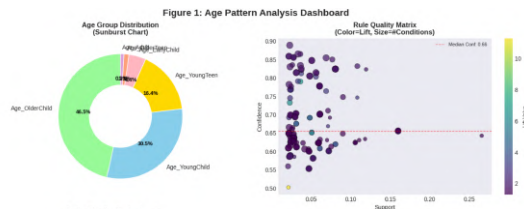


Fig. 14: Age Distribution and Rule Quality

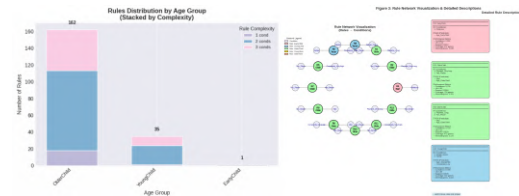


Fig. 15: Rule Distribution by Age and Rule Networks

The 6-binned discretization using pattern mining generated insights into age-related patterns. Discretizing data more efficiently could result into better predictions.

Tree Structure of Rules and Top Five Interesting Rules generated by 6-binned dataset are as follows:

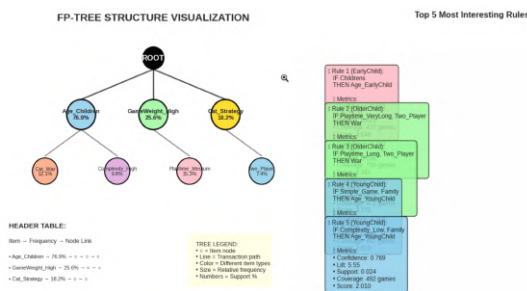


Fig. 16: Final Pattern Mining Results

B) Discretization Techniques:

We evaluated multiple discretization strategies, including Equal-Width, Equal-Frequency, and Natural Binning (KMeans). Silhouette scores were compared across these techniques to identify the most effective binning approach.

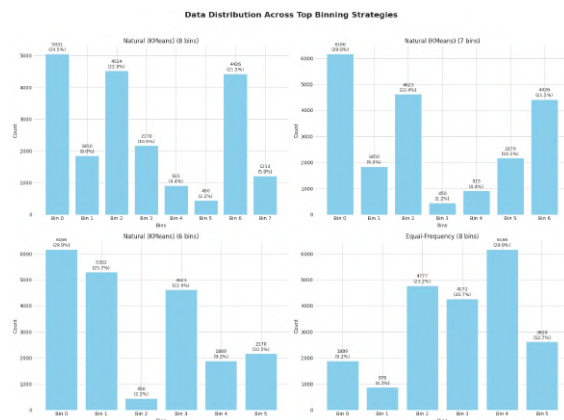


Fig. 17: Comparison of discretization techniques

Overall, KMeans-based discretization consistently yielded higher silhouette scores across different bin configurations.

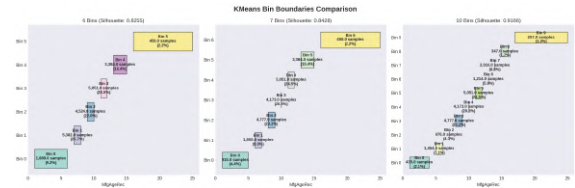


Fig. 18: KMeans discretization with 6, 7, and 10 bins

Resultantly, Natural (KMeans) Binning with 10 bins achieved the highest silhouette score (0.9166). Since this analysis focuses on predictive analysis, silhouette performance across bin counts was considered a more reliable selection criterion.

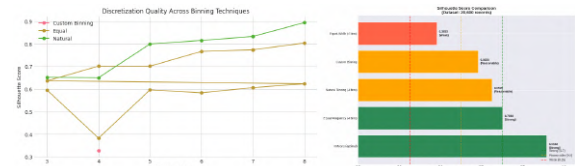


Fig. 19: Silhouette comparison across discretization strategies with respect to bin count

The KMeans discretization with 10 bins was passed with FP-Growth algorithm to analyze association rules, this time association rules were generated for almost all age-groups with majority in (11-12 age).

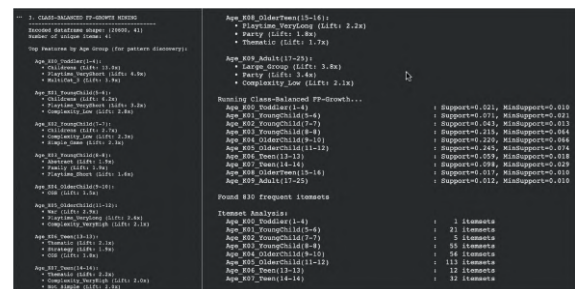


Fig. 20: Association rules generated using FP-Growth with KMeans-based discretization

Using FP Growth, a total of 830 frequent itemsets were identified. Associations revealed clear progression in game characteristics with age: younger groups associated with children-oriented, low-complexity and short-playtime games; whereas older children showed strong association with war, strategy, thematic and high-complexity games. Adult clusters were linked to party games, large group and longer playtimes, although very small in size.

Almost 73 high-confidence rules were filtered. The strongest rule highlighted that two-player games and very long playtime were predictive of

war-themed with high complexity within 11-12 age group, having confidence of 0.77 and lift exceeding 7.

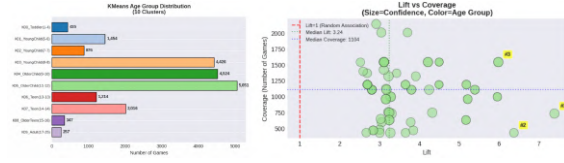


Fig. 21: AgeGroup Distribution and Coverage of Games wrt Lift

One Rule satisfying one age-group each with different confidence, lift and coverability are shown below:

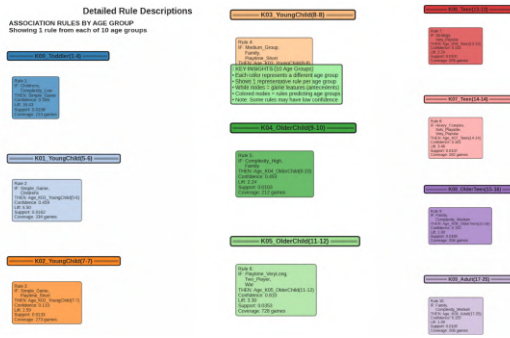


Fig. 22: Rule, Confidence and Lift along each KN-Based Age Group Bin

Missing MfgAgeRec was imputed using KMeans discretization and FP Growth but results indicate limited predictive effectiveness.

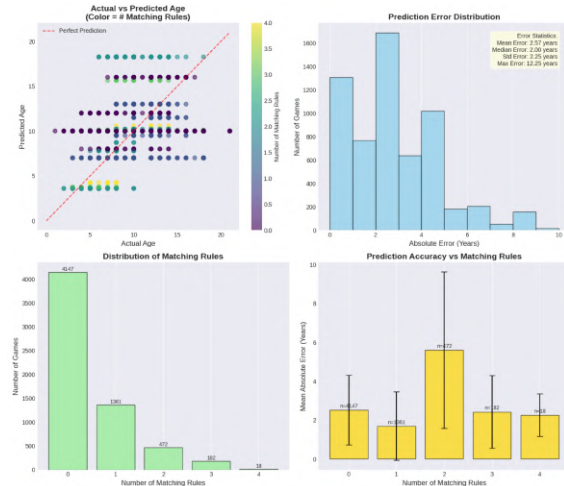


Fig. 23: Performance of Rule-Based Filling, Less Matching Rules in Missing Data

We achieved Mean Absolute Error (MAE) of 2.57 and RootMeanSquared Error (RMSE) of 3.42 while negative R^2 0.52, which confirms that rule-based predictions performed worse than simple mean-median approaches. The overall accuracy remains insufficient for reliable imputation.

The **major contributing factors** are:

- A) **Low Rule Coverage:** Approximately 67% of games had no matching association rules coverage at all.
- B) **Binning-Strategy Mismatch:** Although KMeans Binning is sound with high silhouette score, its integration with FP-Growth requires repetitive patterns. Such patterns were not present at this level of discretization, making the method unsuitable as a standalone solution.
- C) **Data Sparsity After Discretization:** Increasing the number of age bins fragmented the dataset across multiple categories, reducing support for frequent itemsets and weakening the relevance of the generated rules.

Experimental Analysis

To examine the method further, we tried the 2nd best KMeans Binning with 8 bins and we implemented KMeans, Decision Trees, Simple Median Mode and FP Growth with 8 Bins to compare the results.

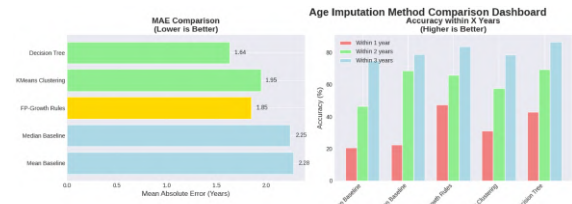


Fig. 24: MAE and Accuracy Comparison in KMeans, Decision Trees, FPGrowth and Baselines

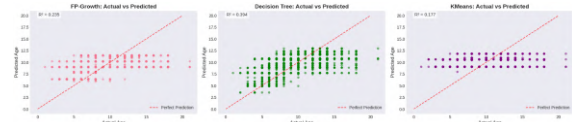


Fig. 25: Actual v.s. Predicted for all three techniques

Decision Tree is the most effective out of all the strategies. Hence Decision Tree was used to impute the Missing values for MfgAgeRec.

ComAgeRec. This feature contained approximately 5530 missing values, and several records were expressed as decimals. With decimals carried on, we performed different analysis. Since the feature resembles *MfgAgeRec*, we tried similar approaches to fill missing values.

Baseline Methods for filling missing values: Mean and Median performed poorly.

For FP-Growth we first discretized data using KMeans. KMeans provided silhouette up to 10 bins (highest silhouette at 10), but considering previous insights from *MfgAgeRec*, we selected 8 bins based on the Silhouette Graph Curve.

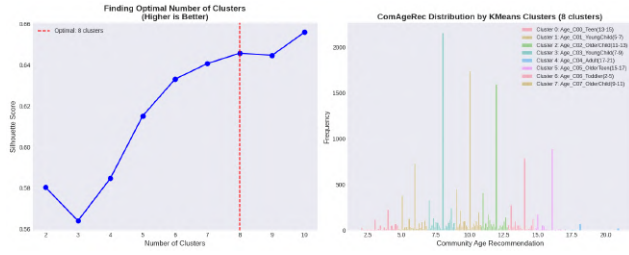


Fig. 26: Silhouette Scores Curve with respect to KMeans classes and visualization of all 8 clusters generated by KMeans.

FP-Growth showed some improvement over baseline, but rules generated only covered 7.6% of the games data.

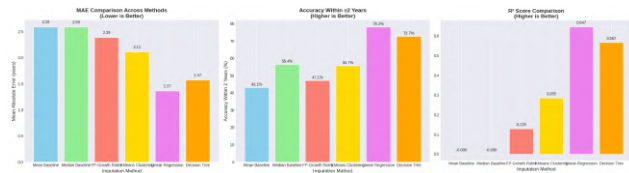


Fig. 27: Rules Confidence and Rules Support for all 8 clusters, along with top rules for each cluster (i.e., rules with higher confidence).

KMeans Clustering ($K = 8$) performed better than FP-Growth, reducing error, improving accuracy, and increasing R^2 , but still not sufficient.

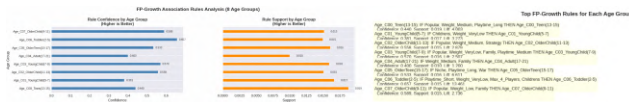


Fig. 28: Bar chart representing R^2 scores, MAE, and Accuracy of all strategies.

Supervised Learning techniques outperformed all other strategies. Linear Regression (using features with correlation ≥ 0.2 achieved comparable MAE and R^2 as using all numerical features; using all features improved overall prediction accuracy) topped all strategies with the highest R^2 and lowest MAE & RMSE. Decision Trees were second best after Linear Regression.

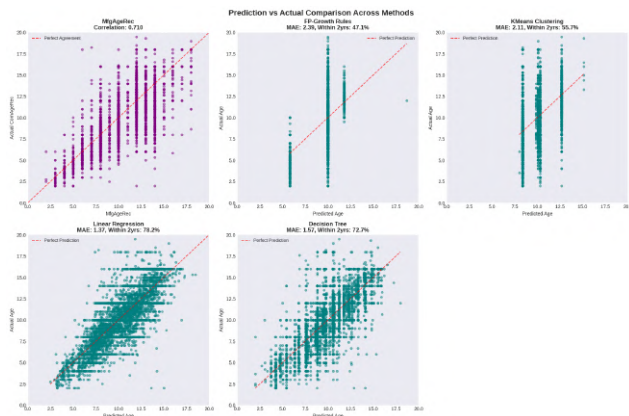


Fig. 29: Prediction versus Actual Results of all different strategies.

Linear Regression was selected to impute missing values. Although FP-Growth works well in theory, due to limited coverage of rules, it was not suitable for this dataset.

MinPlayers and MaxPlayers. Both variables contained default zeros that do not represent valid player counts (50 for *MinPlayers*, 173 for *MaxPlayers* and 132 for *MaxPlayers* < *MinPlayers*). Since the number of players is inherently discrete, mean imputation was inappropriate. We tried KNN with $K=5$ and Random Forest Regressor to fill missing values but Median and Mode values (2 for *MinPlayers*, 4 for *MaxPlayers*) was selected as the most representative replacement for default values.

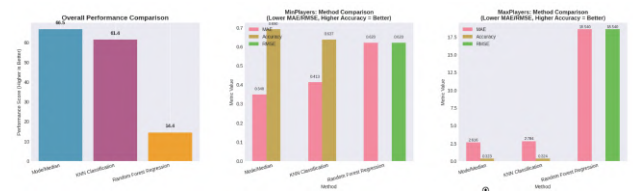


Fig. 30: MAE and RMSE comparison for MinMax-Player using Mode-Median, KNN ($K=5$), and Random Forest

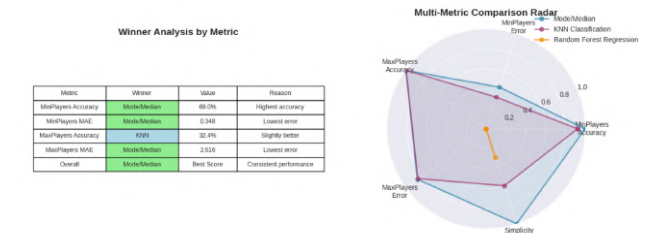


Fig. 31: Mode-Median as Optimum for MinMaxPlayer prediction

For better data distribution, missing values are imputed using a **Hybrid approach**:

- Mode and median values are used to fill missing values in *MinPlayers*, while KNN imputation is applied to *MaxPlayers*.
- If the *BestPlayers* column exists, then *MaxPlayers* is constrained to be greater than or equal to *BestPlayers*.
- If *MinPlayers* exceeds *MaxPlayers*, the values are swapped to maintain logical consistency.

Results were transformed to Logarithmic Scale so that we can have better visuals. (i.e 21000 original data and 4500+ imputations very big ratio)

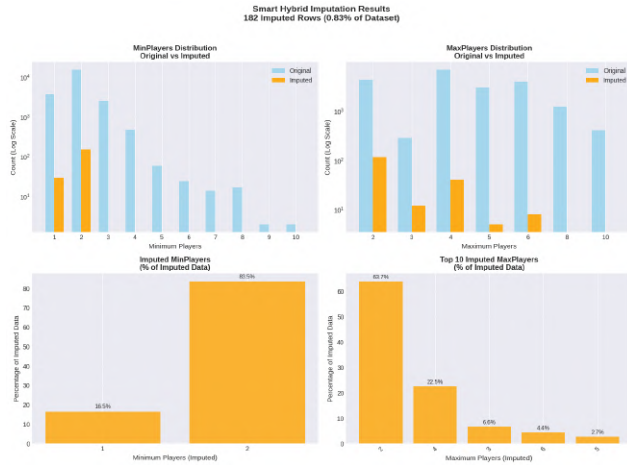
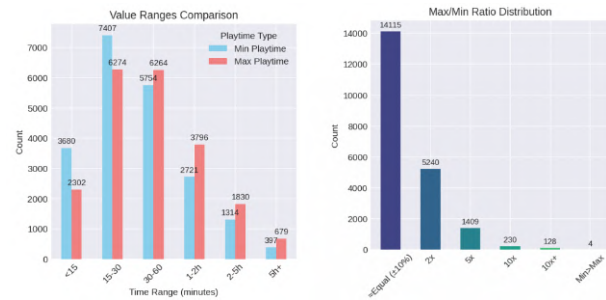


Fig. 32: MinMax Players Missing Value Filled Using Hybrid Approach (Log Transformation)

Popularity Metrics: NumOwned, NumWant, NumWish. These features can have extreme low values (zeros) and extreme high values (potential outliers or based on popularity) but that's the nature of these features, hence those were kept as those are.

Playtime Variables: ComMinPlaytime, ComMaxPlaytime and MfgPlaytime These attributes contained numerous default zeros (780 for *MfgPlaytime* and *ComMaxPlaytime*; 652 for *ComMinPlaytime*); and 633 rows where both were zero. Four entries where $\text{ComMinPlaytime} > \text{ComMaxPlaytime}$ (such values were swapped to make it logical).



The dataset after Cleaning contained 37 numeric features (Idea was to not lose data while Filling Missing Values) along with that Rating was converted to Numerical hence making overall 38 Numerical Features. PCA was applied to identify principal components capturing majority of variance.

17 Components capture 90% of total Variance. PC1 (19.76%), PC2 (14.26%), PC3 (6.1%), PC4(5.9%) and PC5(5.5%).

3.1 Curse of Dimensionality

Within 21,925 rows and 41 features, original dataset was not affected by curse of dimensionality. PCA further reduced features to 17 improving efficiency for other components of project.

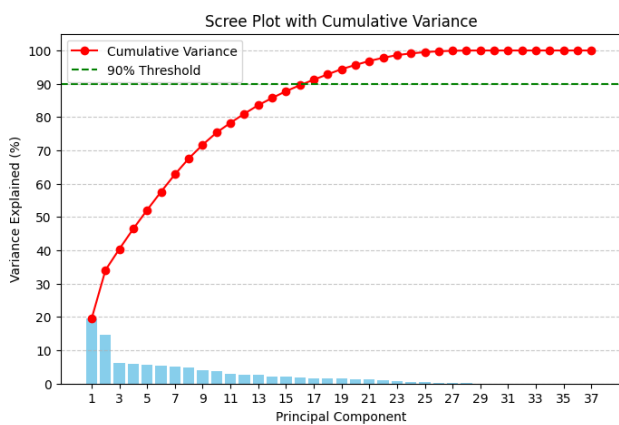


Fig. 36: Principal Component Analysis and Variance

PCA was applied and saved in separate variable for seeing Curse of Dimensionality. Clustering however, is performed on raw transformed data before PCA.

4 Clustering

4.1 Analysis by centroid-based methods

KMeans Clustering was performed on all numerical columns (Non-PCA) and results were:

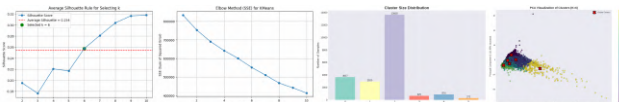


Fig. 37: Selection of K for Clustering using SSE and Average Silhouette along with Representation

K value was selected based on SSE elbow and also confirming this by taking average of all silhouette scores and selecting the average silhouette based K-Value. Both results were same. But we also tried K=8 which had better silhouette and results were:

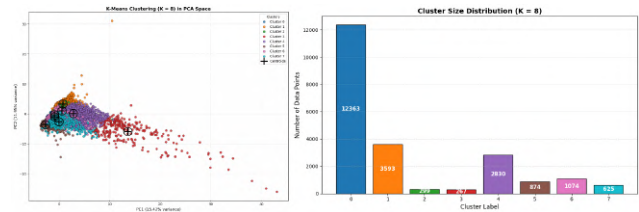


Fig. 38: K=8 Centroid based Clustering with representation

Clearly First Cluster contains almost half of the data. To have more idea we performed Clustering with K=4. All results combined as below:

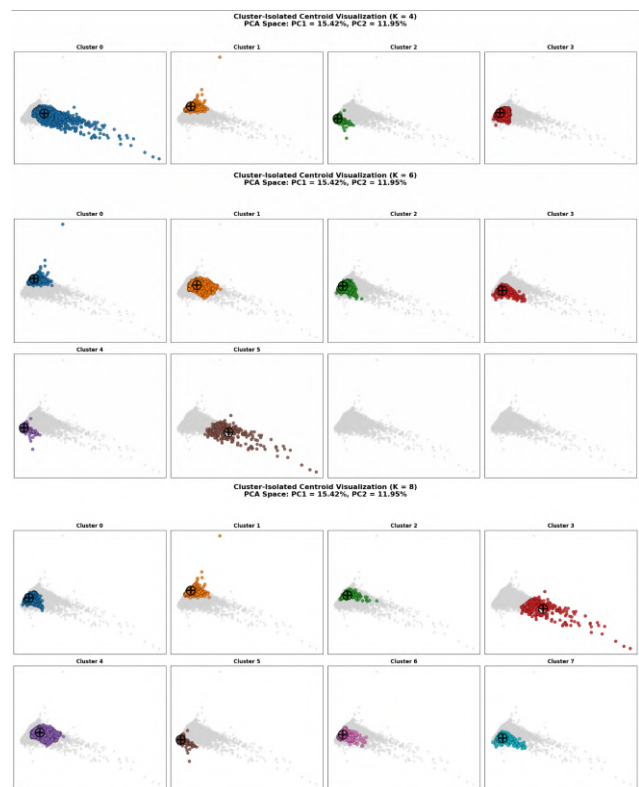


Fig. 39: Centroid based Clustering with K=4,6,8 with representation

PC1 and PC2 explain 22.37% of the variance, Although PCA was used for visualisation after Clustering. Clustering was performed on transformed features, hence there was no loss of information during model fitting.

4.1.1 Bisecting K-Means and X-Means

XMeans was performed to obtain optimal value of K. The algorithm compared silhouette scores and suggested K=8 having best silhouette score so far. Although when Bisecting KMeans was performed Silhouette scores were different compared to Centroid based KMeans.

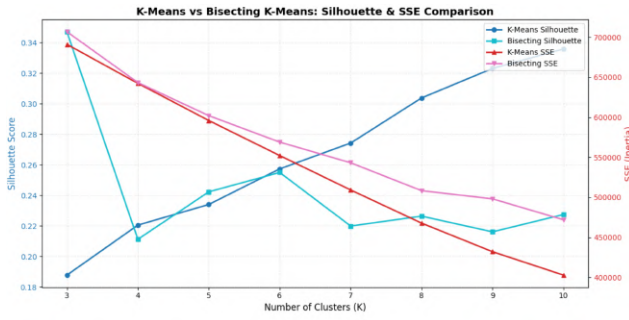


Fig. 40: Silhouette Comparison of KMeans and Bisecting KMeans

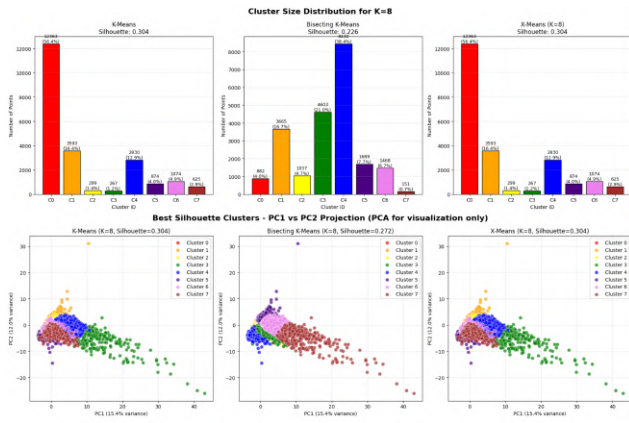


Fig. 41: Distribution of Data wrt KMeans, Bisecting, XMeans

Since XMeans finds best K value and then does simple KMeans, it is obvious to have same results for KMeans and XMeans, although changes are seen with BiSecting KMeans despite same K value.

4.2 Analysis by density-based clustering

4.2.1 DBSCAN

DBSCAN was applied on dataset with MinPts parameter selected as twice the features count hence 76. Epsilon was selected using K-Distance graph, resulting 1.68326. DBSCAN was able to identify 8 clusters.

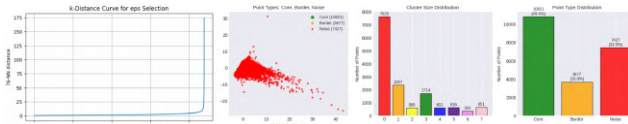


Fig. 42: EPS Selection, 8Clusters by DBSCAN and Division of Rows in Core, Border and Noise Points

In initial run, approximately 33.9% of data points (7427) were classified as Noise. To check if those points could form more Clusters, DBSCAN was reapplied on the Noise Points. For the second iteration, We increased epsilon to 1.8, 1.9 and 2.0. 2.0 generated better results along with MinPts set to 38 (Original Feature Size).

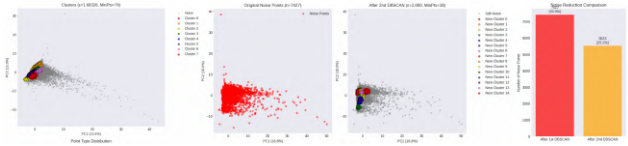


Fig. 43: 1st Iteration Points then Reduction of Noise Points in 2nd Iteration

Noise points reduced from 7427 to 5523 and clusters increased from 8 to 16. This tells that iterative execution of DBSCAN can identify additional weaker structures within the dataset.

4.2.2 (OPTICS)

Continuing from DBSCAN results, OPTICS was applied to explore more on density-based Clustering. To maintain consistency with previous DBSCAN analysis, we used similar hyperparameters; MinDistance = 30, Euclidean distance was selected as points are far away and our data is transformed, cluster-method = 'xi' as we already performed DBSCAN, xi value was checked between 0.05 and 0.07, 0.07 gave better results. Reachability plot generated is as under:

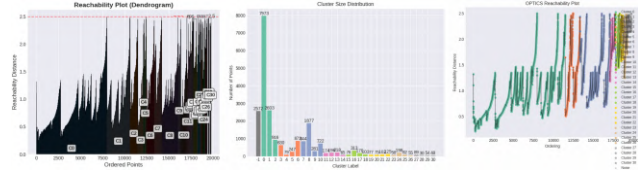


Fig. 44: Reachability Plot, Clusters by Reachability Valleys and Distribution

Only 2572 points were classified as Noise Points, which is better than DBSCAN with 2nd Iteration although number of Clusters here increased to 31.

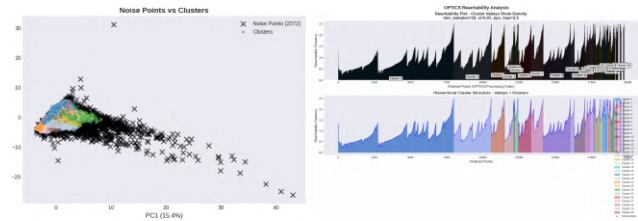


Fig. 45: Noise Points v.s. Clusters along with Density Valleys for Clusters Formation

The reachability plot shows dense valleys capturing structures in the data.

4.3 Analysis by Hierarchical Clustering

Using the entire dataset of 21,925 rows exceeded 12GB of RAM, hence we used a 50% stratified sample (obtained using KMeans with K=8). The stratified sample has an average difference of 0.04% and a maximum difference of 0.15%.

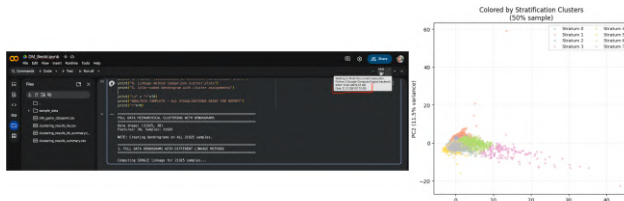


Fig. 46: 12GB RAM usage on full data and 50% stratified distribution

Hierarchical clustering was performed via Single Linkage, Complete Linkage, Average Linkage, and Ward Linkage.

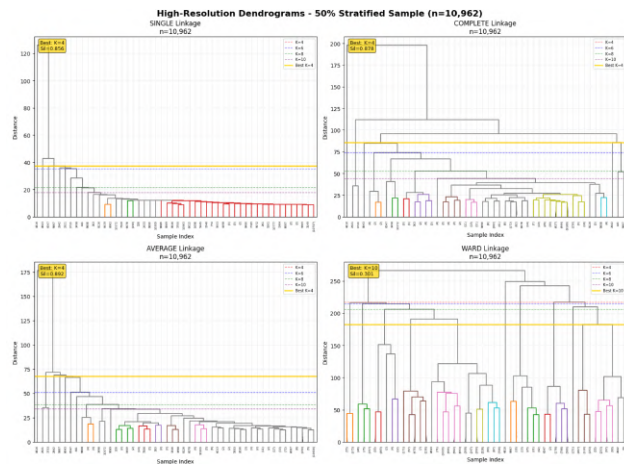


Fig. 47: Dendograms with different linkage methods

Top performers were Average Linkage with $K = 4, 5, 6$. The best silhouette was 0.892 with $K = 4$.

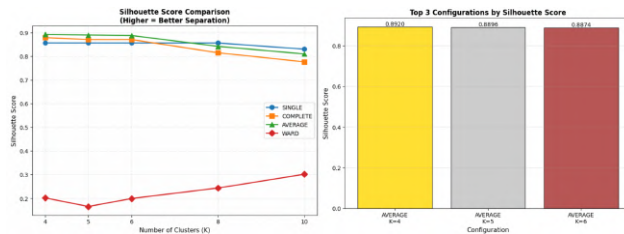


Fig. 48: Best linkage: all Average Linkage configurations

Clearly in Dendograms, Average Linkage despite greater Silhouette has mostly data distributed in only one Cluster and Ward Linkage having lesser silhouette score distributed data in better way. Hierarchical Clustering distribution based on different Linkage methods plotted are shown as:

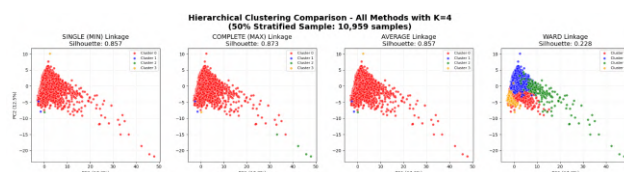


Fig. 49: Ward Linkage Getting Better Distribution than Others (Whole Dataset 21925 Rows)

4.4 Hierarchical Density-Based Clustering

Final test was conducted using H-DBScan; hyper-parameters selected were from DBScan and OPTICS' verdict. MinCluster size = 30 (), Euclidean Distance, ClusterSelection Epsilon = 3. MinSamples = 15. Some parameters defined were calculated by multiple execution of H-DBScan.

H-DBScan resulted in 18 Clusters with only 1133 Noise Points. Membership probability to each data point was also calculated which shows 97.8% probability that the points belong to right cluster.

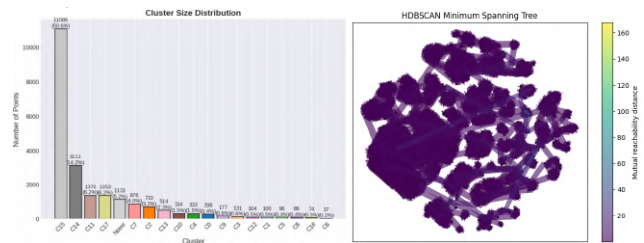


Fig. 50: Clusters Distribution and Minimum Spanning Tree

Due to internal implementation of HDBScan for large datasets and our limited computing capacity, we were not able to perform full-dataset dendrogram visualization; therefore stratified sample after HDBScan (accumulating almost all clusters) was performed for visual representations.

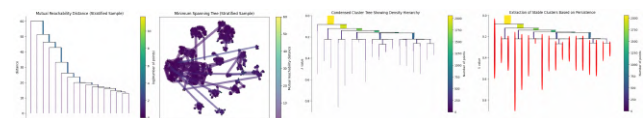


Fig. 51: Mutual Reachability, Minimum Spanning Tree, Condense Cluster Tree and Extraction of Stable Clusters on Stratified Sample

This confirms that H-DBScan clusters are highly stable, making results more effective compared to DBScan and OPTICS. Although number of clusters is still bit large.

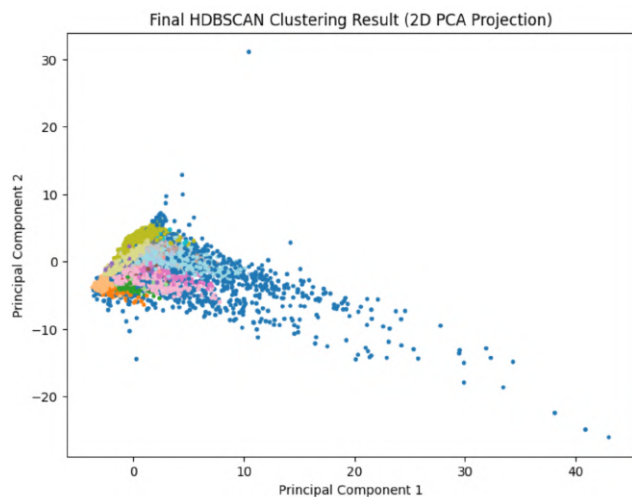


Fig. 52: Cluster Distribution using HDBScan

5 Classification

For Classification, we must have a target variable; seeing our dataset Rating column is perfect fit for a target. Current it has three classes: Low, Medium and High which are encoded to (1,2 and 3). We are using all numerical features as input features to predict Rating class.

5.1 K-Nearest Neighbours

Applying **K-Nearest Neighbors** Classification. For obtaining best K value, we tried Elbow Plot (K=6 best), Stratified Cross Validation (K=8 best) and K-Fold (10 Folds) Cross Validation Strategy (K=10 best).

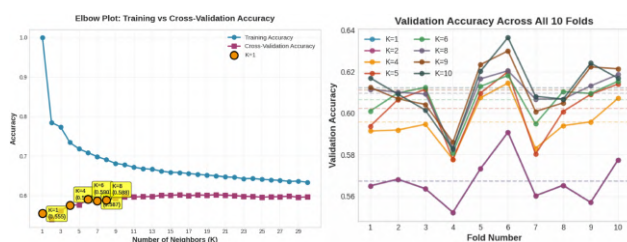


Fig. 53: K-Nearest Neighbors Elbow Plot, Cross Validation and K-Fold Results



Fig. 54: Performance of KNN with K=10

Results of KNN (K=10) on Stratified validation set are better than rest of K values (1, 6, 8, 148). K=1 and K=148 (Square root of Sample size) were discarded to avoid Overfitting and Underfitting.

5.2 Naive Bayes Classification

For Naive Bayes, Gaussian Naive Bayes was selected as all features are numerical. Although the results with Naive Bayes were not good compared to KNN.

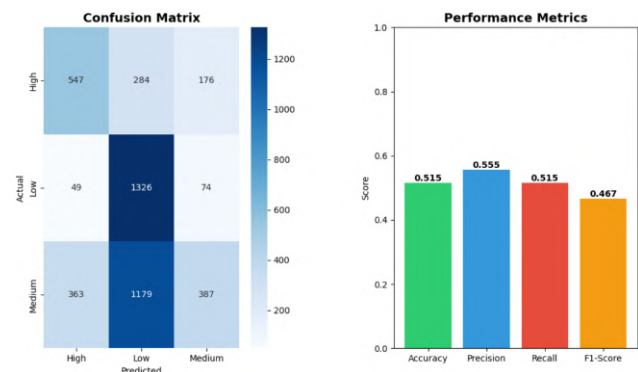


Fig. 55: Performance of Naive Bayes

We tried to convert Continuous Features into Discrete by Binning, in order to apply another variant of Naive Bayes, but the results were even less.

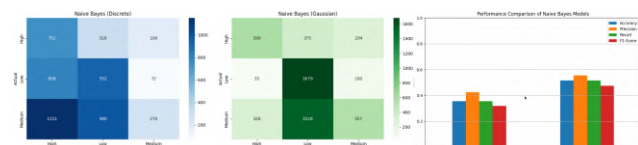


Fig. 56: Naive Bayes on Binned Data and Gaussian

Naive Bayes fails to distinguish between target classes by high margin than KNN (High Recall (0.92) for Low rating and poor Recall (0.20) for Medium), leading to poor probability boundaries.

5.3 Decision Trees

For Decision Trees, first Tree was induced from original non-transformed dataset. Nodes are being split using both Gini Index and Entropy. First three levels of the trees are displayed below. Both impurity measures show similar top levels. Hence for easy computation, Gini was selected for further analysis.

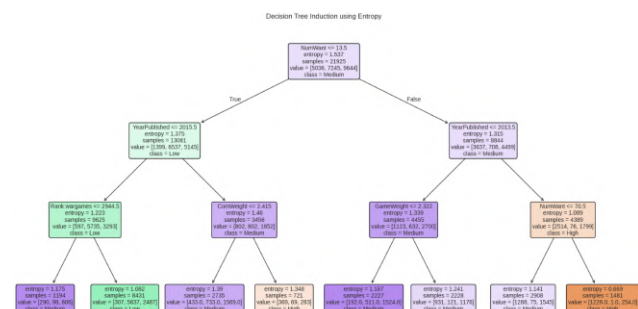


Fig. 57: Induction of Tree using Entropy

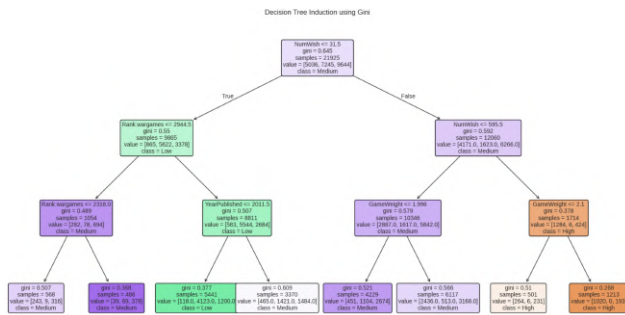


Fig. 58: Induction of Tree using GINI Index

Results, however were better than Naive Bayes and KNN.

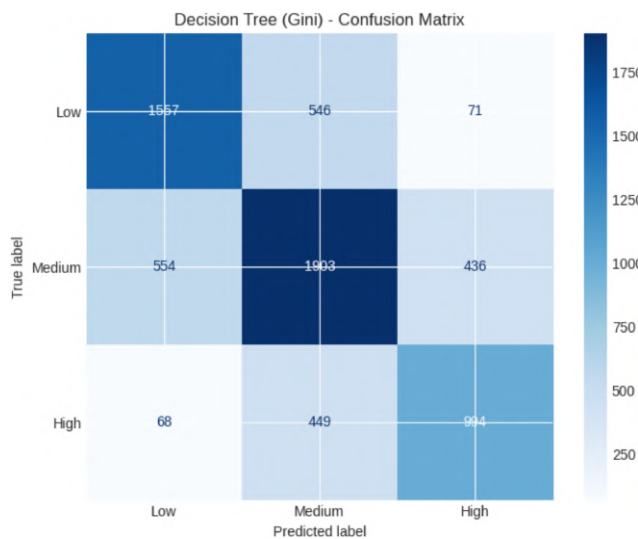


Fig. 59: Confusion Matrix using GINI Index

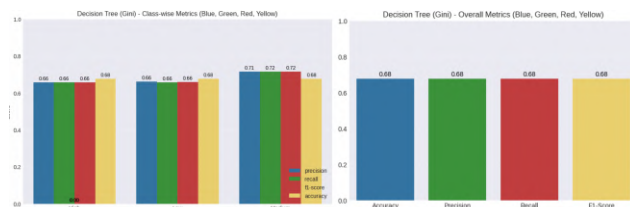


Fig. 60: Accuracy, Precision, Recall and F1-Score Overall and Class-wise

We also checked transformation before applying Decision Tree induction but results were little improved (+0.1 to 0.3 difference).

All three models were compared using ROC Curves under different misclassification costs (Accuracy let alone may not always define value).

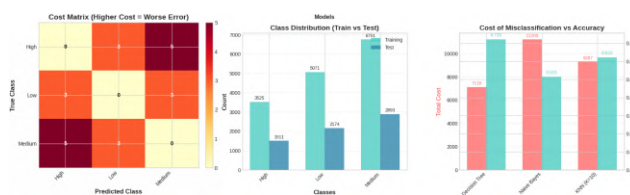


Fig. 61: Cost Matrix, Train Test Distribution, Accuracy v.s. Misclassification Cost

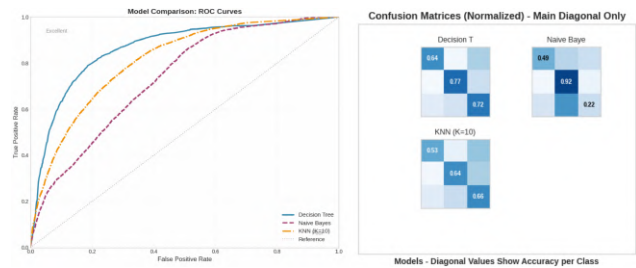


Fig. 62: ROC Curves and Confusion Matrices of Accuracy per Class

With Cost Misclassification added in ROC Curve, Better curve means fewer costly mistakes. Decision Trees outperformed all other models in terms of Accuracy and Cost Situation. The similar trend was observed in Filling Missing Values in starting chapters. *Here Transformed Data is used with Decision Trees (as it gave little better results)*

5.3.1 Further Analysis on Categorical Features

Each Categorical Feature was used to make a Decision Tree model to classify a category (as there exists games belonging to multiple categories). At first we removed rows where a Game was belonging to none of available categories (i-e having all Cat: features as 0). Left-over data 10741 rows were used then. All Category features were used as Target variables, we removed 'Rank:columns to avoid data leakage, rest of numerical features were used as Input. Dataset was divided Train and Test splits and Decision Trees were formed.

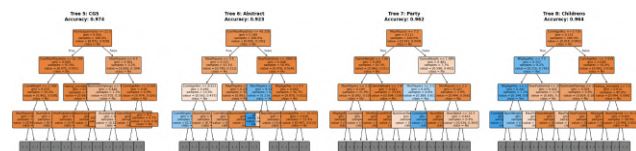


Fig. 63: GINI Split Trees for Top Categories

To check multiple categories association of a game, we check all categorical models and see their output (1 if belongs to category else 0).

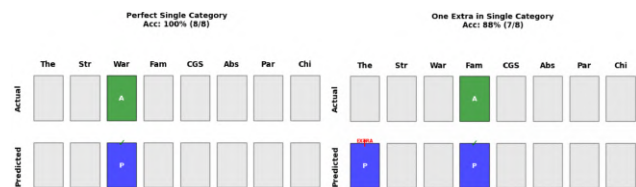


Fig. 64: Binary Array to represent Multi Categories of a Row



Fig. 65: Performance Metrics of Categories and Accuracy wrt Multi-Category

Classification with One Category or Two Categories is good, three combined would be better but due to less samples it is not sufficient.

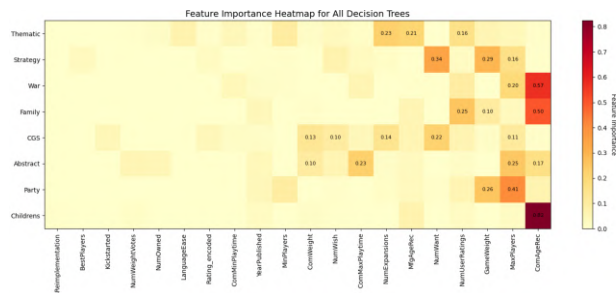


Fig. 66: Heatmap of Important Features wrt Categories

However, overall accuracy of Model predicting exact category is less 0.54 on Test Data.

5.4 Regression

Same Categorical Classification was performed using Regression Model (Simple LR and Lasso) and the results were:

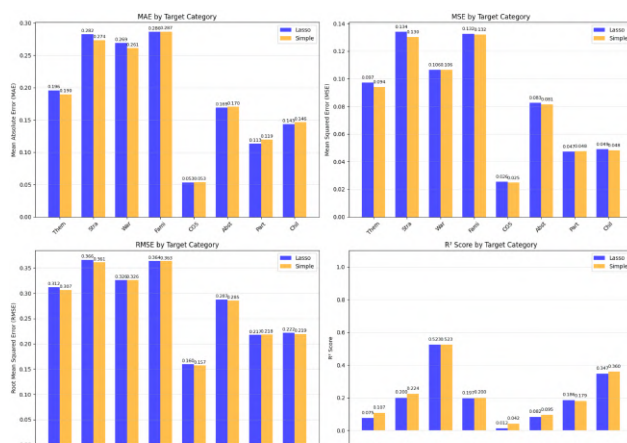


Fig. 67: Class-Wise Mean Absolute Error, Mean-SquaredError, RootMeanSquaredError and R^2

Residual Plots drawn are meaningless as values are Binary. All points will be just straight lines over 0 and 1. We tried Mapping each class with 1-8 numbers and plotting points but again with Discrete values the plots were still vague.

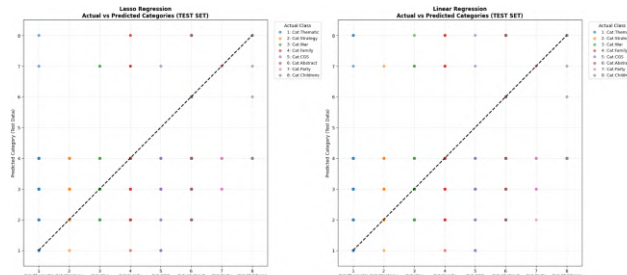


Fig. 68: Approximation of Residual Plot Lasso v.s. Simple LR

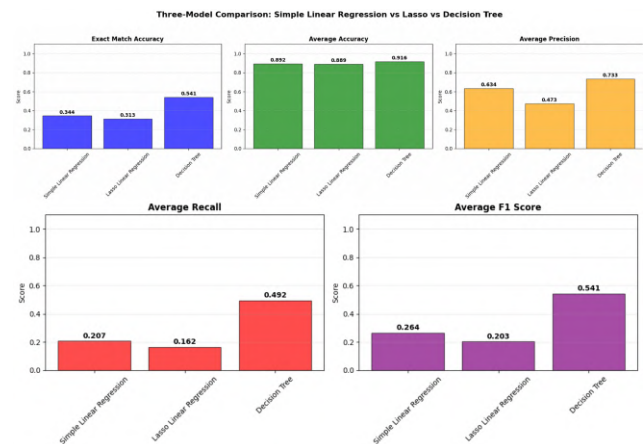


Fig. 69: Comparison of LR, LR Lasso and Decision Tree Performance

Testing **One Target Variable and One Independent Variable** Seeing the correlation we have highest correlation of NumOwned with NumUserRatings. Applying Linear Regression wit NumOwned being Target.

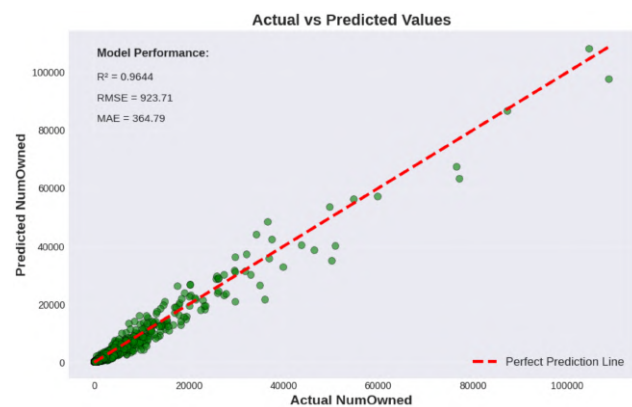


Fig. 70: Residual Plot for NumOwned along with Performance Metrics

We can see perfect prediction with smaller error rate but at the same time MSE is in thousands, which does not mean high error but presence of Outliers. Clearly NumOwned is a popularity feature hence it could have outliers and it is treated normal.

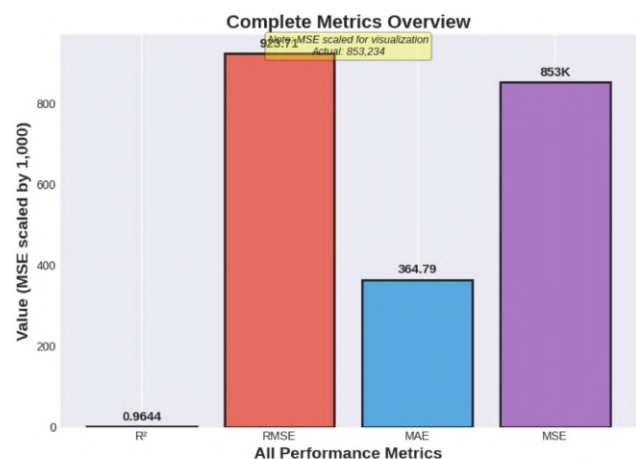


Fig. 71: Different Performance Metrics with MSE scaled

Again Testing **One Target Variable and Multi Independent Variable** This time more columns were added such as: NumUserRatings, NumWeightVotes, NumWish and NumWant. All these features have correlation > 0.5 . The results were

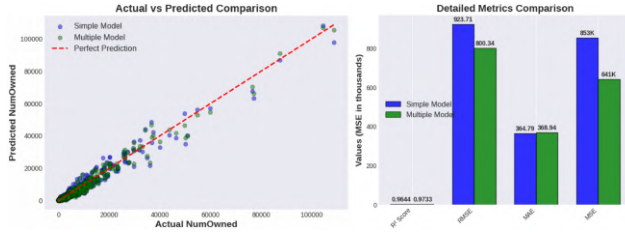


Fig. 72: Residual Plot and Different Performance Metrics for Single and Multi Independent Features

Multiple Feature based Regression performed better, there is slight increase in predictive power but RMSE and MSE reduced by big margin although there was very slight increase in MAE (4.15 units) but still multi model well captures the complexity of ownership prediction.

Despite Linear Regression's continuous target nature, we performed Rating classification as well just to experiment. All numerical features were used for LR and the results were:

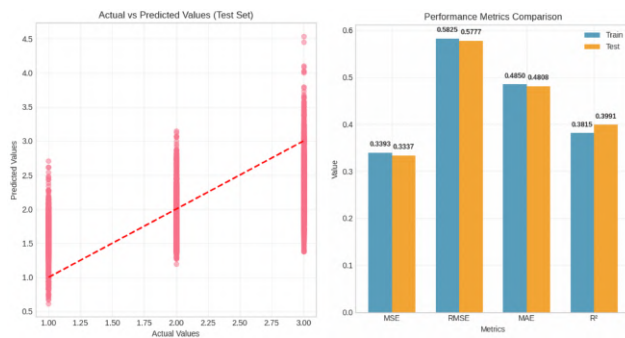


Fig. 73: Residual Plot and Different Performance Metrics for Rating case

Rating feature has encoded values (1,2 and 3). An RMSE of 0.5777 shows that: on average, the predictions differ by 0.58, which is half a distance from completely another class. Hence averagely predictions would be off by half a class. Similarly MAE is 0.48 again almost half class distance.

6 Pattern Mining

We already saw working of Pattern Mining while dealing Missing Values, here we will see in more detail.

6.1 Frequent Pattern Extraction:

We preprocessed the data, dropped useless features (Ranks, Name, Family), negligible quantity features (GoodPlayers and BestPlayers) and highly correlated features (ComWeight having 0.997 correlation with

GameWeight, NumWish having 0.94 correlation with NumWant; although these two features were interpreted as different in initial analysis; NumUserRatings having corr of 0.985 with NumOwned and 0.917 with NumWeightVotes). Then created Bins for Numerical (Non Binary or Tertiary) Features using qcut and cut with some customisations by hit and trial.

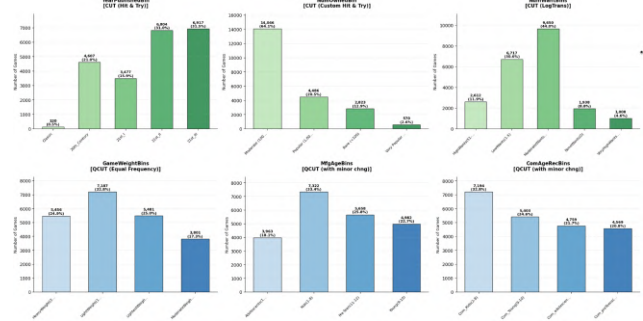


Fig. 74: Distribution of different features after Binning

Using **Apriori**, we set min-support 20 and z-min 2 (minimum number of items in itemset) to generate Frequent Itemsets. All feature values were transformed into transactions to generate itemset using Apriori.

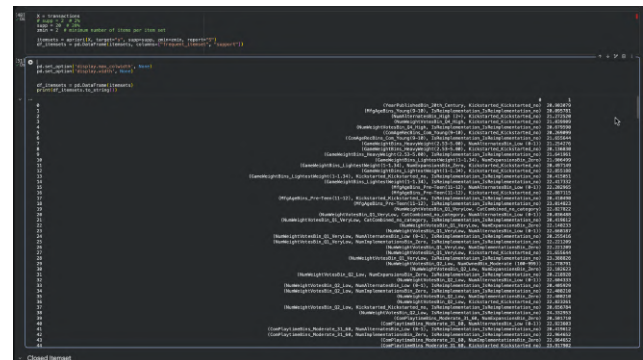


Fig. 75: Sneakpeak of Frequent Itemsets

The top three frequent itemsets based on above configurations show that “Games belonging to 20th Century are not Kickstarted”, “Games of Mfg Age group 9-10 years are not ReImplemented” and “Games having Alternates ≥ 2 are also not Kickstarted”. These Itemsets might not be much interesting but with defining confidence and lift, one can see interesting rules.

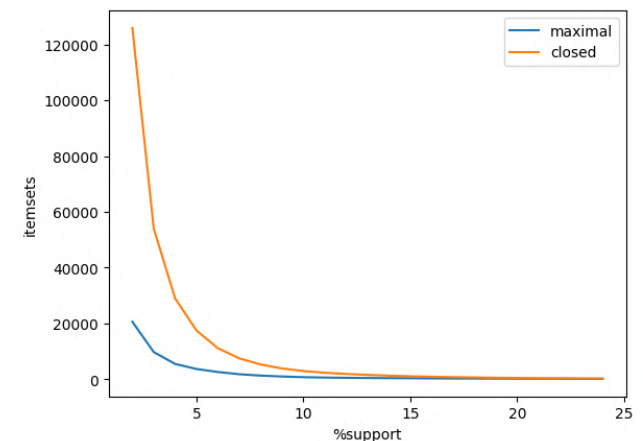


Fig. 76: Support Bar Itemset vs Support

We changed minimum support to 2% and z-min to 3 pairs. **Qualitative Observations:** Frequent Itemsets with 2% support reveal features with absences mostly such as: if a game is reimplemented, kickstarted, has zero expansions. So we tried seeing tail of the itemset which showed some interesting patterns.

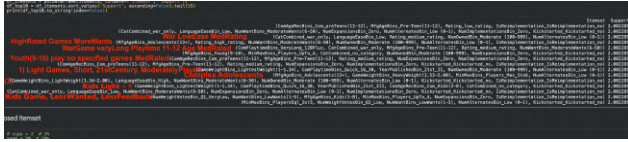


Fig. 77: Frequent Itemsets 2% Support (Least Patterns)

Itemset tagged with “War LowEase MedRating” shows such games exists only 2% yet they have Medium Rating and still 100-999 Ownership. Moderate Complexity War Games are Reliable Niche market, predictable demand steady income.

“Complex Adolescents(13+)” Heavy Weight games with 5-6 max players are popular among Adolescents - One could drive market demand from here. “High Rated games More Wanted”, Adolescent (13+) games with High Ratings are more wanted, shows demands. “Youth(9-10) Playful” shows Young audience (9-10) play games (upto 4 players count) with no specified category achieve moderate ownership. “Lightweight 2” shows Lightweight games with High LanguageEase achieve moderate ownership and demand without crowdfunding - self sellers.

The number of Frequent Patterns decrease as min-support increases. High support shows dominant trends while low support shows niche areas. 2% support produced rare but insightful patterns while high support filters out noise and miss specialised combinations.

To see better Frequent Itemsets, we removed some binary features like NumExpansions, NumAlternates e.t.c. and results now for min-support 2% and z-min 2 were:



Fig. 78: Frequent Itemsets 2% Support (After Dropped Cols)

Showed sense-making frequent patterns now. Using this new transaction dataset for Association Rules generation.

6.2 Association Rules Extraction:

First Applied Apriori with confidence of 0.6 and minimum 3 items to see associations (29619 Rules Generated).

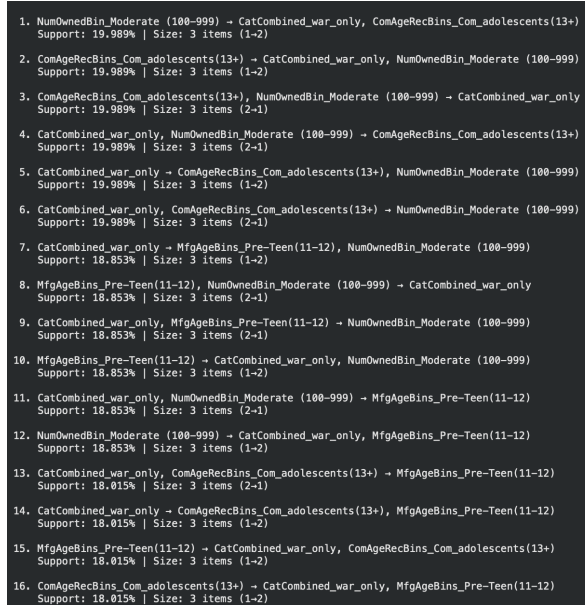


Fig. 79: Association Rules with 60% Confidence

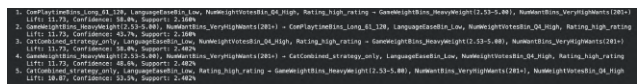


Fig. 80: Association Rules with High Lift

Interpretations:

A If a board game has long playtime (61-120 mins), Low LanguageEase, Many WeightVotes and High Rating; It is 11.7 times more likely than random to also have Heavy Weight in High Demand (High NumWants).

B In Strategy Category Games, Complex LanguageEase, many WeightVotes and High Rating are likely to have more GameWeight and high sought-after (High Lift).

C Children Games (2-8 years old) with Low Rating, published in 20th Century, with less GameWeight are 10.6 times more likely to belong to Children Category (High Confidence) <Could be Data Leakage here as Cat:childrens and ComAgeRec both are taken>

D HeavyWeight Games and High Demand games with Long Playtime, Complex LanguageEase and High Rating are bidirectional associations with lift of 11.73. (High Lift) <Information could be used to produce more Games in this niche>

E Adolescent targeted Strategy games with Moderate LanguageEase are published in 21st Century.

The confidence was changed to 0.4 and results were (65756 Rules Generated):

```

1. NumOwnedBin_Moderate (100-999) → CatCombined_war_only, ComAgeRecBins_Com_adolescents(13+)
Support: 19.989% | Size: 3 items (1-2)
2. ComAgeRecBins_Com_adolescents(13+) → CatCombined_war_only, NumOwnedBin_Moderate (100-999)
Support: 19.989% | Size: 3 items (1-2)
3. ComAgeRecBins_Com_adolescents(13+) → NumOwnedBin_Moderate (100-999) → CatCombined_war_only
Support: 19.989% | Size: 3 items (2-1)
4. CatCombined_war_only, NumOwnedBin_Moderate (100-999) → ComAgeRecBins_Com_adolescents(13+)
Support: 19.989% | Size: 3 items (2-1)
5. CatCombined_war_only → ComAgeRecBins_Com_adolescents(13+), NumOwnedBin_Moderate (100-999)
Support: 19.989% | Size: 3 items (1-2)
6. CatCombined_war_only, ComAgeRecBins_Com_adolescents(13+) → NumOwnedBin_Moderate (100-999)
Support: 19.989% | Size: 3 items (2-1)
7. CatCombined_war_only → MfgAgeBins_Pre-Teen(11-12), NumOwnedBin_Moderate (100-999)
Support: 18.853% | Size: 3 items (1-2)
8. MfgAgeBins_Pre-Teen(11-12), NumOwnedBin_Moderate (100-999) → CatCombined_war_only
Support: 18.853% | Size: 3 items (2-1)
9. CatCombined_war_only, MfgAgeBins_Pre-Teen(11-12) → NumOwnedBin_Moderate (100-999)
Support: 18.853% | Size: 3 items (2-1)
10. MfgAgeBins_Pre-Teen(11-12) → CatCombined_war_only, NumOwnedBin_Moderate (100-999)
Support: 18.853% | Size: 3 items (1-2)
11. CatCombined_war_only, NumOwnedBin_Moderate (100-999) → MfgAgeBins_Pre-Teen(11-12)
Support: 18.853% | Size: 3 items (2-1)
12. NumOwnedBin_Moderate (100-999) → CatCombined_war_only, MfgAgeBins_Pre-Teen(11-12)
Support: 18.853% | Size: 3 items (2-1)
13. CatCombined_war_only, ComAgeRecBins_Com_adolescents(13+) → MfgAgeBins_Pre-Teen(11-12)
Support: 18.015% | Size: 3 items (2-1)
14. CatCombined_war_only → ComAgeRecBins_Com_adolescents(13+), MfgAgeBins_Pre-Teen(11-12)
Support: 18.015% | Size: 3 items (1-2)
15. MfgAgeBins_Pre-Teen(11-12) → CatCombined_war_only, ComAgeRecBins_Com_adolescents(13+)
Support: 18.015% | Size: 3 items (1-2)

```

Fig. 81: Association Rules with 40% Confidence

The wargame category has bidirectional associations with PreTeen (11-12) and Adolescent (13+) age group and moderate ownerships. One can take insights from this rule and create games targeting this age audience. Confidence Increased to 0.80. (10741 Rules Generated)

```

1. Rule1:
NumOwnedBin_Very Popular (10,000+), Rating_high_rating → NumWantedBin_VeryHigh(20+), NumLightnessBin_04_High
Lift: 18.37
Confidence: 93.5%
Support: 12.306%
Size: 4 items (1-2)
2. Rule2:
ComplexLanguageBin_Low, NumOwnedBin_Very Popular (10,000+), Rating_high_rating → NumWantedBin_VeryHigh(20+), NumLightnessBin_04_High
Lift: 18.38
Confidence: 93.3%
Support: 12.306%
Size: 5 items (1-2)
3. Rule3:
ComAgeRecBins_Com_adolescents(13+), NumWantedBin_VeryHigh(20+) → GameWeightBin_HeavyWeight(12.53-5.88), NumLightnessBin_04_High, Rating_high_rating
Lift: 8.22
Confidence: 87.9%
Support: 12.306%
Size: 5 items (1-2)
4. Rule4:
CatCombined_children_only, GameWeightBin_LightnessBin(1-1.34), NumWantedBin_VeryHigh(20+) → ComAgeRecBins_Com_Kids(2-8), MfgAgeBins_Kids(1-8), Rating_low_rating
Lift: 8.47
Confidence: 88.7%
Support: 12.306%
Size: 6 items (1-2)
5. Rule5:
CatCombined_children_only, NumOwnedBin_Moderate (100-999), NumWantedBin_VeryHigh(20+) → ComAgeRecBins_Com_Kids(2-8), MfgAgeBins_Kids(1-8), Rating_low_rating
Lift: 8.47
Confidence: 88.7%
Support: 12.306%
Size: 6 items (1-2)
6. Rule6:
CatCombined_children_only, NumWantedBin_VeryHigh(20+) → ComAgeRecBins_Com_Kids(2-8), MfgAgeBins_Kids(1-8), Rating_low_rating
Lift: 8.22
Confidence: 87.9%
Support: 12.306%
Size: 5 items (1-2)
7. Rule7:
CatCombined_children_only, GameWeightBin_LightnessBin(1-1.34), YearPublishedBin_20th_Century → ComAgeRecBins_Com_Kids(2-8), MfgAgeBins_Kids(1-8), Rating_low_rating
Lift: 8.22
Confidence: 88.6%
Support: 12.306%
Size: 6 items (1-2)

```

Fig. 82: Association Rules with 80% Confidence

Rule1 in 80% Confidence shows, If a Game is very popular (10,000+ NumOwned) and Rated High then almost 93.5% confidence that it also has Very High Demand (NumWanted) and many Votes.

Rule2 illustrates that adding Complex Language in above Rule barely changes outcome (still 93.3% confidence)

Rule3, again Adolescent market, this rule can generate insight that Age 13+ is threshold where games can develop complexity and still be high rated.

Finally, Confidence Increased to 0.90 yielded 3551 Rules. The number of Rules decrease as Confidence increases. Although each Confidence level was targeting a different depth, Higher confidence generated categorical associations, lower confidence generated rules based on Ownership, Simplicity and Weights.

6.3 FP-Growth Rules Extraction:

Using FP-Growth with confidence 0.4, we obtained consequent mostly for Category:Party and Category:Children with lift of 13.34 and 11.23

Fig. 83: FP Growth Rules with 40% Confidence

With confidence of 0.6, we obtained rules mostly with consequent Category:Children games with High lifts: ()

Fig. 84: FP Growth Rules with 60% Confidence

While Apriori dominated in identifying patterns with numerical/bin features (Ownership, Weight, LanguageEase, Ratings), FP Growth with 40% confidence uncovers high lift rules for Categorical Features specifically Children and Party categories. FP Growth with 40% confidence is good candidate to finding Missing Values in Categorical Features. Trying to Exploit Categorical Rules to Fill Missing Values. Although an attempt was already made in Data Understanding Chapter but with different Feature.

	rules_count	avg_confidence	max_confidence	avg_lift	max_lift
CatCombined					
CatCombined_strategy_only	3550	0.560939	1.000000	2.005976	8.435992
CatCombined_childrens_only	1232	0.624870	1.000000	2.637562	5.720398
CatCombined_war_only	15791	0.640746	1.000000	1.937030	3.846512
CatCombined_family_only	2444	0.537942	0.932735	1.748453	3.972095
CatCombined_family_strategy_true	75	0.572980	0.927928	1.936722	3.551091
CatCombined_thematic_only	186	0.497969	0.847390	1.665852	4.441862
CatCombined_abstract_only	521	0.527837	0.834101	1.774083	6.301474
CatCombined_strategy_thematic_true	11	0.516475	0.764192	1.622500	2.395631
CatCombined_party_only	48	0.547165	0.751055	2.284606	7.942823
CatCombined_cgs_only	8	0.484833	0.610879	1.453759	2.030126

Fig. 85: Rules for Categories within 40% Confidence Extraction

Despite having lots of Rules, We used Top-1 FP Growth rule to see if it matches category correctly. With one it performed good.

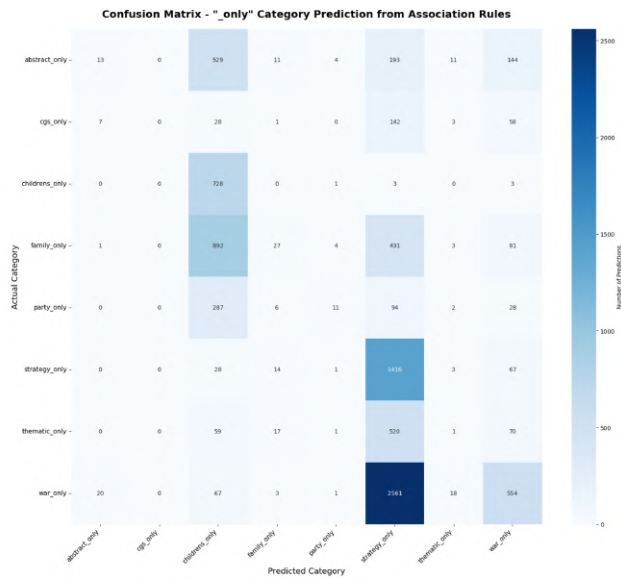


Fig. 86: Confusion Matrix to Predict Category by FP Rules(1Rule/Cat)

The above confusion matrix is only for categories where exists only one category for a game. Then we tried with 3 Rules per Category and 3 Unique/Distinct Rules per Category to see results.

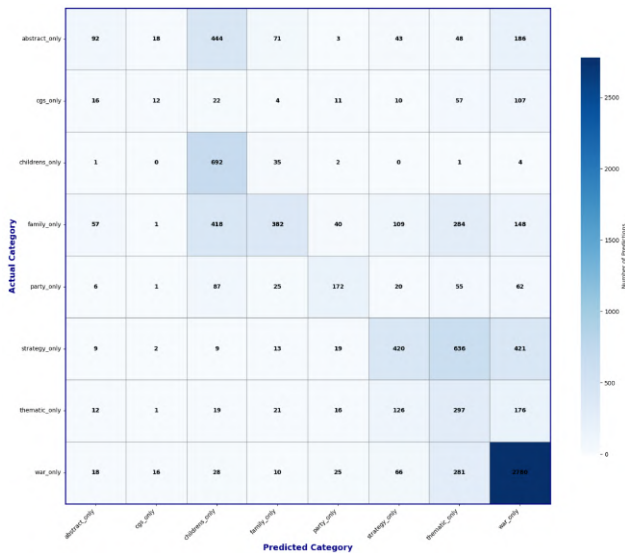


Fig. 87: Confusion Matrix with Top 3 and Unique 3 FP Rules

As we had a lot of Rules for WarGames, It looks like FPGrowth has performed very well in predicting Category War. With increasing number of Rules to see prediction, results are getting better. We tried to apply all Rules to see final results, it took more than **40minutes** to execute.

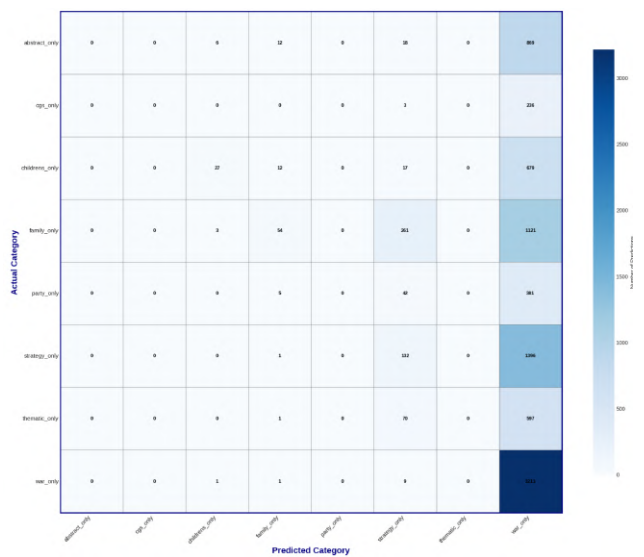


Fig. 88: Confusion Matrix with All Categorical Rules

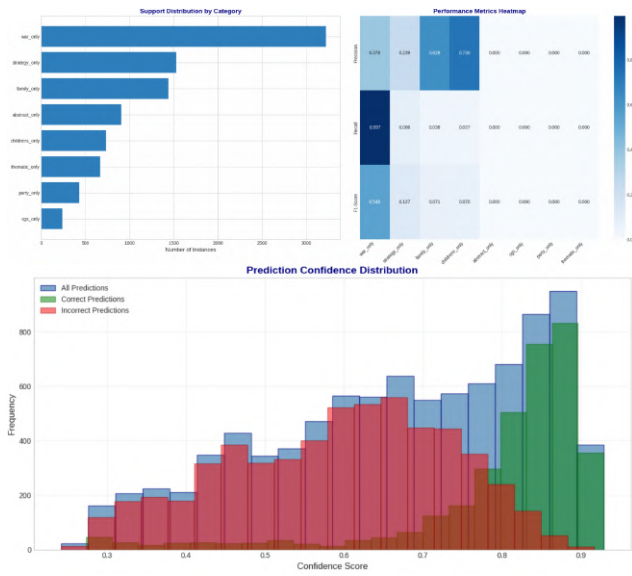


Fig. 89: Support Distribution, Performance and Overall Predictions

Results after using all Categorical Rules were more biased to one Category having the most Rules. Certainly better Rules number per category can be selected and Pattern Mining can be used to achieve far more better results but it takes a lot of time to hit and try on all rules.

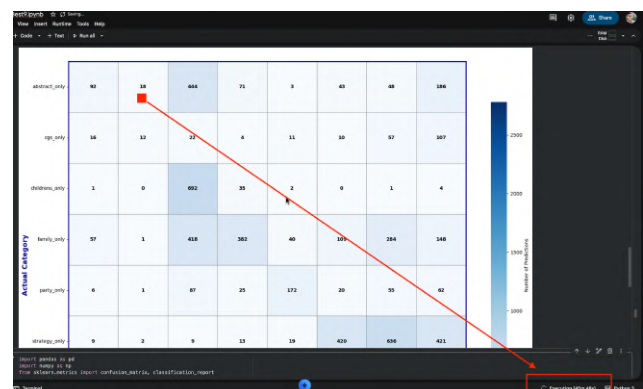


Fig. 90: Execution of All Rules Time Complexity

In above chapter, it shown to be less effective when compared to other techniques (Classification, Regression, Normal Mean). However, When data is highly categorical, FP Growth can be of great help in filling Missing Values.

Summary

The project conductive complete data mining analysis on BoardGame dataset (21925 games, 41 features) through preprocessing, clustering, classification, regression and pattern mining. The initial idea was to drop as less data as possible, hence different strategies were observed and then implemented carefully to handle missing data in such a way. Data Proprocessing chapter in itself covered almost at chapters alone. Legitimate outliers were preserved to understand sparse and diverse nature of dataset. Overall, Dimensionality was less as Principal Component Analysis (PCA) reduced features to 17 PCs covering 90% of variance. Clustering was performed before PCA but used results for visualizations. Decision Trees almost outperformed other algorithms in classification as well as missing values. Pattern Mining and FP-Growth revealed strong associations which could be used to drive business-insights. Different algorithms are tested using different performance metrics throughout chapter. Some Experiments and Hinge testings were also conducted throughout different chapters. The processed dataset maintains board game diversity while providing foundation for predictive modelling with Decision Trees and Linear Regression proving most effective.

Finished

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